

SafeDecontam: Risk-Controlled Benchmark Decontamination

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Abstract—Benchmark corpus filtering requires an explicit operating point: permissive rules leave leakage in place, whereas aggressive rules delete legitimate related material. SafeDecontam is a source-level deletion rule applied after an upstream retriever returns candidate documents for known benchmark items. Its evidence adapter combines lexical similarity, constraint agreement, answer support, chronology, and page-level pooling; the scorer is fixed before calibration. For each source item, calibration applies an exact one-sided tolerance procedure to the largest score among retain-labeled candidates, and the method abstains when the clean calibration reservoir cannot support the requested collateral-deletion budget. We evaluate SafeDecontam on constructed pairs from public benchmark artifacts and on raw web-search evidence; in a blinded and adjudicated Massive Multitask Language Understanding (MMLU) audit, it achieves 81.71 percent recall, 100 percent precision, and 0.9894 area under the receiver operating characteristic curve (AUROC), with no clean removals among 71 items (one-sided 95 percent exact upper bound 4.13 percent). Recall drops on external benchmark families and a cross-lingual probe, so evidence coverage and calibration data must be established for each family. SafeDecontam is thus a corpus-deletion policy for retrieved candidates with an explicit damage budget and abstention condition; it does not establish model training exposure or score inflation.

Index Terms—benchmark contamination, corpus filtering, risk control, data governance, large language models

I. INTRODUCTION

Open benchmarks support reproducible model comparisons, but the same accessibility exposes questions, worked solutions, and minor variants to data-collection pipelines [1], and evaluation scores may then mix generalization with prior exposure even when a prompt does not reproduce stored text verbatim [2]. Contamination studies usually begin after training: protected or transformed evaluations test robustness to a changed surface [3], and retrieval audits locate likely source matches [4]. Corpus curation instead asks what can be removed before training and at what cost. Literal overlap is a poor boundary here: rephrasing or translation may preserve decisive content while changing nearly every local string [5], while tutorials, earlier sources, and same-topic questions share vocabulary without revealing the answer.

The central problem is threshold selection under asymmetric error costs: a permissive cutoff misses derivatives, an aggressive cutoff removes more legitimate material, and pairwise AUROC ranks candidate records but neither specifies a tolerable deletion risk nor reveals how false removals spread across benchmark source items during corpus-wide screening.

SafeDecontam formulates the decision over source groups: a replaceable scorer maps the adapter’s evidence to a removal score, the largest retain score in each group calibrates a one-sided non-parametric tolerance threshold once validation fixes the scorer, and a cutoff is returned only when clean groups support the requested budget.

Experiments cover a constructed pool from three public benchmark ecosystems and released MMLU search results, with source groups kept intact across partitions. This paper contributes (i) a source-level definition of collateral deletion, where a source is damaged if any legitimate related candidate is removed; (ii) an auditable evidence adapter and an exact clean-only calibration rule that meets a pre-specified damage budget or abstains; and (iii) a group-preserving evaluation with a separated raw-search protocol whose final MMLU decisions are checked by blinded human adjudication.

A. Scope and Threat Model

SafeDecontam operates after an upstream retriever has produced candidate documents for a known benchmark item, and considers accidental or non-adaptive benchmark reuse, including exact copies, paraphrases, translations, and answer-bearing fragments. It neither guarantees recall for candidates missed by retrieval nor defends against an adaptive adversary optimizing content against the frozen adapter or threshold. The protected asset is legitimate corpus material; the controlled loss is the probability that at least one retain-labeled candidate of a source item is deleted. The guarantee assumes a fixed scorer, evidence schema, and labeling rule, with clean groups drawn from the same within-family distribution as the calibration reservoir; label noise, a change of benchmark family, language, or retrieval source, or a changed parser or encoder voids it, and Section III-E states how a reservoir is built for each family.

II. RELATED WORK

Contamination audits operate on two surfaces: corpus-side methods search for source material, whereas model-side probes measure behavior consistent with prior exposure. Shingle containment offers an inexpensive exact-match baseline [6], and Okapi Best Matching 25 (BM25) adds term weighting and query normalization [7]. Both remain lexical: neither encodes answer preservation, chronology, or structured numerical changes, and their evidence weakens as a derivative moves away from literal reuse.

Risk-controlling prediction sets supply the statistical template [8]: grouping retain candidates by source and calibrating their maximum score aligns the calibration variable with the

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TABLE I
CONSTRUCTED PAIR POOL BY FAMILY.
GSM8K: GRADE SCHOOL MATH 8K; MCQ: MULTIPLE CHOICE.

Family	Groups	Remove	Retain
Arithmetic (GSM8K)	1,319	5,272	6,873
MCQ (MMLU)	505	2,000	1,515
MATH Level 5	253	506	1,012
Total	2,077	7,778	9,400

deletion event of interest, and a one-sided tolerance limit on these maxima controls source-group damage directly.

III. METHOD

A. Decision Unit and Losses

For benchmark item b and candidate document d , let $s(b, d)$ denote an increasing removal score. All candidates associated with b form source group g ; grouping makes the deletion event explicit, since damage occurs when the policy removes any legitimate material linked to the same benchmark item.

The remove class contains exact copies, verified derivatives such as released rephrasings [5] and answer-preserving mathematical variants [9], and answer-bearing fragments. The retain class includes changed-answer perturbations even when they remain close to the source on the surface [10], together with same-topic items, predating sources, and unrelated controls.

At threshold τ , pair contamination recall (CR) and pair collateral deletion rate (CDR) are

$$\text{CR} = \Pr[s \geq \tau \mid Y = 1], \quad \text{CDR} = \Pr[s \geq \tau \mid Y = 0]. \quad (1)$$

A source item incurs damage when any of its legitimate candidates is deleted; group collateral deletion rate (GCDR) measures the fraction of source groups with at least one removed retain candidate. CR and CDR summarize pairwise behavior; GCDR captures whether a small pairwise error is dispersed across many benchmark items at deployment scale. In the constructed pool, CR and CDR are pair-level and GCDR is group-level; in the raw-search audit, each source item forms one group after page-level pooling, so reported recall and clean-removal rates are source-item quantities, not the pair-level rates in (1).

B. Observable Evidence and Score Models

Six similarity features form the lexical layer. Four-gram containment and three-shingle Jaccard measure local reuse [6]; query-normalized BM25 supplies term-weighted retrieval [7]. Word- and character-level term frequency-inverse document frequency (TF-IDF) cosines capture broader overlap, while a 64-dimensional latent-semantic cosine provides a low-rank view; vocabulary construction and projection fitting are confined to the training partition.

Structured MATH perturbations show why lexical similarity cannot determine whether a changed number reverses the answer [11]. The adapter records numbers, relation cues, target tokens, entities, and units; counts answer support only

in conclusion-like context; checks the first four available constraints; and uses chronology only with a reliable predating source [12].

Candidate scorers include each individual feature, their mean, similarity-plus-chronology and full-evidence logistic models, and a 16-unit multilayer perceptron (MLP) with rectified-linear-unit activation, quasi-Newton optimization, and ℓ_2 regularization; validation selects one scorer before calibration begins. MLP thresholds use the unsquashed margin, which avoids boundary ties from saturated logits. The protocol separates fitting, calibration on clean-group maxima, and one-time evaluation under the locked decision rule.

C. Clean-Only Exact Calibration

For family h , let R_{hj} be the retain-labeled candidates of calibration source group j and $Q_{hj} = \max_{i \in R_{hj}} s_i$ their largest score; n_h counts the calibration groups with a nonempty retain set; group j is damaged exactly when this maximum crosses the threshold. For k allowed exceedances, the one-sided upper bound is

$$U_h(k) = F_{\text{Beta}}^{-1}(1 - \alpha/H; k + 1, n_h - k), \quad (2)$$

where H is the number of calibrated families: $U_h(k)$ is the largest per-group damage probability that remains consistent, at confidence $1 - \alpha/H$, with observing k exceedances among n_h clean groups, and Clopper-Pearson inversion makes it exact in finite samples [13]. Set $k_h^* = \max\{k : U_h(k) \leq \varepsilon\}$. For $Q_{h,(1)} \leq \dots \leq Q_{h,(n_h)}$, τ_h is the smallest representable value above $Q_{h,(n_h - k_h^*)}$, so tied maxima remain retained. The shared rule uses $\tau = \max_h \tau_h$; an empty feasible set returns no threshold.

Coverage claim. Fix the score model before calibration and assume that clean-group maxima within each family are independent draws from that family's distribution, which future groups also follow, as in [8]. With $R_h(t) = \Pr(Q_h \geq t)$ denoting the probability that threshold t damages a clean group,

$$\Pr\left(\max_{h \leq H} R_h(\tau) \leq \varepsilon\right) \geq 1 - \alpha. \quad (3)$$

A threshold above $Q_{h,(n_h - k)}$ leaves at most k observed maxima in the upper tail; under continuous scores that tail mass follows the Beta order-statistic law, so k_h^* supplies coverage $1 - \alpha/H$ within family h , and the largest family threshold with a union bound yields (3); retaining tied maxima keeps the bound conservative for discrete scores.

D. Abstention and Evidence Semantics

Abstention is an explicit output: SafeDecontam withholds a threshold when the retain reservoir is too small, the evidence schema is unvalidated for the family, or no exceedance allowance meets the requested budget; a numerical cutoff would otherwise overstate the calibration support. Missing or weak metadata cannot become affirmative evidence by default in this policy.

E. Calibration Reservoirs in Practice

A reservoir is collected per family, with families prespecified by benchmark domain, language, and retrieval source before any score is inspected. Its source items are sampled from the intended deployment stream and pass through the production retrieval depth, parser, and pooling policy; retain relations are adjudicated from full pages where available, derivatives stay with their source, and reservoir items are disjoint from fitting, validation, and test groups. Since $k = 0$ is feasible only when $n_h \geq \log(\alpha/H)/\log(1-\varepsilon)$, at least 59 clean groups are needed for $\varepsilon = \alpha = 0.05$ and one family (80 per family for three), and smaller reservoirs abstain. A new language, retrieval source, or pipeline version receives a new frozen version and reservoir.

IV. EXPERIMENTAL DESIGN

A. Public-Data-Derived Pair Construction

Table I summarizes the constructed pool: arithmetic groups originate from GSM8K items and their released solutions [14], the multiple-choice groups draw on three MMLU subjects [15], and the Level-5 MATH subset contributes the third family [16]. Source identity determines the grouping key, while the relation between source and candidate determines the remove or retain label for subsequent evaluation.

The remove class contains aligned released rephrasings, answer-preserving GSM-Plus variants used in the later sensitivity analysis, MMLU symbol-replacement substitutions, and solution-only or answer-bearing fragments, since a record can expose an answer without reproducing the full question. Retain relations are harder than random negatives: answer-changing GSM-Plus and GSM-Symbolic variants preserve wording but require a different result, MATH-Perturb adds two levels of structured variants, and simple arithmetic word-problem items [17], predating Academia Sinica Diverse math-word-problem items, Logical Question Answering (LogiQA) [18], and Truthful Question Answering (TruthfulQA) [19] supply same-topic, chronology, reasoning, and cross-domain controls.

The pool contains public artifacts and deterministic pairs, not an independent crawl: a three-shingle rule links 253 of 279 MATH-Perturb Level-5 records (best match at least 0.35 and 1.5 times the runner-up), failed matches remain unassigned, and four empty records are removed before feature extraction.

Table II makes the labeling rule explicit: preservation of decisive benchmark content, not public availability, determines class membership.

B. Partitions, Audit, and Protocol

Within each family, source groups enter deterministic 40/30/30 training, calibration, and test partitions. Derivatives remain with their source. Grouping precedes representation fitting, preventing a transformed item from training a scorer later tested on that source.

Some retain documents pair with multiple source items, so identical text may occur in different partitions without splitting a group.

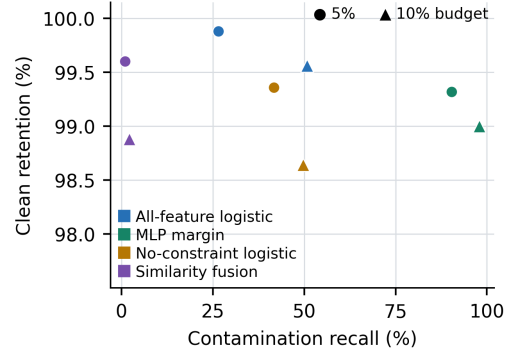


Fig. 1. Calibrated operating points on the constructed pool at 5% and 10% group budgets.

The raw-search audit uses a separate 40/20/20/20 source-item split: training fits representations and scorers, validation selects one scorer, calibration turns retain-group maxima into a threshold for a 5% budget at 95% confidence, and test runs once. Repeated uniform resource locators are removed across splits, and retain assignments, threshold, software, data commit, and 256-bit Secure Hash Algorithm hashes are recorded to enable reproducible audit replay.

C. Integrity Checks and Human Audit

Alignment checks run before feature extraction: each of the 1,319 GSM8K rephrasings must produce the same extracted answer as its source, and four empty questions are discarded [5].

The three MMLU subjects yield 505 source items and 485 nonempty released rephrasings, while the symbol-replacement set matches expected subject-level counts [20]. For MATH, 253 links pass uniqueness, 24 fail a score or separation cutoff, and two lack usable shingles [11].

Human assessment starts after raw-search decisions are stored: two annotators independently label all 154 MMLU test items without predictions or silver labels; at least one opens the full page for 51 items, both for 42, and the remaining 103 rely on released snippets because archived pages are unavailable. A third blinded annotator resolves 71 disagreements, giving 82 contaminated, 71 clean, and one uncertain item. Four-class agreement is 53.90% ($\kappa = 0.365$); among 128 cases for which both annotators are certain, binary agreement reaches 87.50% ($\kappa = 0.727$). The uncertain record is omitted without reassignment after model output is observed during evaluation.

V. RESULTS

A. Constructed-Pair Sensitivity Study

On the constructed arithmetic and MCQ pairs in Table III, the fixed-configuration MLP-margin rerun reaches 90.34% CR, 0.68% pair CDR, and 2.01% GCDR, and ten grouped re-splits yield $86.67 \pm 7.50\%$ recall (population SD; 69.99%–95.03%). Because configuration and threshold selection inspect calibration outcomes, the associated 4.67% calibration bound is descriptive rather than a fresh certificate; Section V-B reports the final protocol-valid evaluation.

TABLE II
OPERATIONAL PAIR CONSTRUCTION: REMOVE AND RETAIN RELATIONS PER FAMILY.

Family	Remove relations	Retain relations
Arithmetic	GSM8K item with solution; released rephrasing; answer-preserving GSM-Plus variant; solution-only answer fragment	Answer-changing GSM-Plus and GSM-Symbolic variants; same-topic arithmetic item; predating math-word item; an unrelated TruthfulQA control
MCQ	MMLU item with answer; released rephrased question; symbol-replacement substitution; answer-bearing options fragment	Different same-subject MMLU item; predating LogiQA item; a cross-domain arithmetic control
MATH	Linked Level-5 MATH item with solution; final-solution fragment	MATH-P-Simple and MATH-P-Hard; different same-type MATH item; random MATH item

TABLE III
EXPLORATORY CONSTRUCTED-PAIR TEST AT THE 5% GROUP BUDGET.

Scorer	CR (%)	CDR (%)	GCDR (%)
4-gram containment	25.15	0.08	0.37
Shingle Jaccard	5.47	0.40	1.65
Normalized BM25	3.72	0.56	2.56
Similarity fusion	0.92	0.40	1.83
No-constraint logistic	41.66	0.64	2.56
All-feature logistic	26.53	0.12	0.55
MLP probability	0.00	0.00	0.00
MLP margin	90.34	0.68	2.01

TABLE IV
PROTOCOL-VALID RAW-SEARCH LOCKED TESTS AT THE 5% BUDGET.

Benchmark	Adapter	Pos./clean	Selected	Recall	Clean	AUROC
MMLU-silver	Full	68/86	Logistic	91.18	5.81	.9518
MMLU-human	Full	82/71	Logistic	81.71	0.00	.9894
Reasoning Challenge	Light	17/223	Logistic	41.18	2.24	.9053
HellaSwag	Light	22/155	Word TF-IDF	63.64	0.00	.9261
C-Eval	Light	129/144	Logistic	46.51	2.08	.9273

Recall and Clean (clean-removal) are source-item-level percentages, not the pair-level rates in (1). Human excludes one uncertain item. Full includes the frozen embedding model; Light uses lexical and constraint evidence.

Recall is 92.21% for arithmetic and 85.43% for MCQ under the shared threshold, and 68.42% for MATH under its local threshold; the smaller MATH reservoir also widens its split range. Raising the prespecified group budget from 5% to 10% moves the margin threshold from 56.40 to 10.67, raising CR to 97.93%, pair CDR to 1.00%, and GCDR to 3.30%; finite deployment budgets remain fixed before test inspection.

Probability ties from saturated logits produce zero recall in four re-splits; unsquashed margins preserve ordering. The 210 false negatives comprise 98 released rephrasings, 63 other transformations, 30 answer fragments, and 19 exact copies (one GSM8K and 18 MMLU). Of 17 false removals, 14 are GSM-Symbolic re-instantiations [10], two answer-changing GSM-Plus items [9], and one MMLU item from the same audited subjects.

Arithmetic and MCQ margin thresholds transfer poorly to MATH [11], removing a retain candidate in 28.95% and 11.84% of source groups, while the MATH-calibrated threshold loses 30.99 and 35.34 recall points on arithmetic and MCQ relative to their local thresholds. Local re-estimation corrects the operating point, not evidence absent from the deployed adapter itself.

B. Protocol-Valid Raw Search Audit

The raw-search audit uses MMLU search responses and silver annotations released by Li et al. [4]. Joining identifiers, discarding empty responses, and deduplicating uniform resource locators leaves 783 items: 467 clean, 58 input-contaminated, and 258 input-and-label-contaminated. Because archived full pages are unavailable, scoring uses result titles and snippets only, and pages later opened by annotators never enter the frozen features; the source-item split assigns 308/159/162/154 items to training, validation, calibration, and test, and silver labels never enter the feature vector.

Validation compares word and character TF-IDF, frozen Beijing Academy of Artificial Intelligence General Embedding small-en-v1.5 cosine [21], full-evidence logistic regression (ℓ_2 penalty, $C = 1$), and a 16-unit MLP (weight decay 10^{-3} , L-BFGS) near 5% empirical clean damage, with concatenated componentwise maxima, top-two means, and overall means of page features for learned scorers; single-feature baselines use maxima. Recall is 53.97%, 49.21%, 31.75%, 79.37%, and 68.25%, and AUROC is 0.8980, 0.8848, 0.8328, 0.9431, and 0.8973, so validation selects logistic regression for later MMLU decisions.

Calibration contains 106 retain groups; at a 5% budget and 95% confidence ($H = 1$), one exceedance is allowed, giving threshold 0.61848 with a one-sided upper bound of 4.397%. The locked test removes 62/68 silver-contaminated items (91.18%; group-bootstrap 95% interval 83.58–97.14%) and 5/86 silver-clean items (5.81%; interval 1.25–11.39%); AUROC is 0.9518. The calibration statement bounds population clean-group risk with 95% confidence over the random calibration sample, not every finite test split, so the observed 5.81% does not contradict the 5% budget.

Human adjudication changes only the evaluation labels; the locked rule removes 67/82 contaminated items (81.71%; exact 95% confidence interval 71.63–89.38%) and 0/71 clean items (one-sided 95% exact upper bound 4.13%). Precision is 100% (confidence interval 94.64–100%) and AUROC is 0.9894; the uncertain item is excluded. With full-page access, removal is 15/20 contaminated and 0/30 clean (one-sided 95% exact upper bound 9.50%). Table IV reports all audits.

Lighter adapters for the Allen Institute for Artificial Intelligence Reasoning Challenge, HellaSwag, and Chinese Evaluation Suite (C-Eval) remove 0–2.24% of clean items but recall only 41.18–63.64%. In the web pilot, removal is 90.74%

for answer-bearing records, 81.82% for question-only records, and 0.00% for cross-item controls. The frozen English-trained MLP-margin probe removes 0/72 exact-text, 8/72 English, and 2/72 Chinese derivatives (4.63% overall) after local calibration; the learned scorer has no exact-match override.

VI. DISCUSSION AND VALIDITY

GSM-Plus and GSM-Symbolic show that broad similarity can survive answer changes, while false negatives cluster in paraphrases and short fragments whose decisive content activates few constraints, so the learned combiner and its score scale remain part of the policy. Threshold transfer fails in the same way: arithmetic or MCQ cutoffs delete substantial legitimate MATH material, and English-trained features miss cross-lingual contamination. Threshold transfer violates the matched-distribution assumption: weighted conformal calibration can bound the coverage loss under bounded drift [22], whereas SafeDecontam requires a matched reservoir per family and abstains otherwise.

Figure 1 separates ranking from policy performance: similar rankings can yield different clean-retention tradeoffs, so policy evaluation reports CR, CDR, and GCDR as decision metrics, retaining AUROC only as a ranking summary [8].

The finite-sample statement bounds population risk at the stated confidence [13], and the 4.13% one-sided upper bound of the adjudicated audit retains sampling uncertainty. Retrieval must favor sensitivity because misses are unrecoverable [7], whereas deletion uses the frozen adapter, scorer, and threshold [8].

Three limitations bound the evidence: MATH-Perturb links use approximate matching [11], and constructed-pool feature comparison inspects calibration outcomes. Raw-search calibration uses silver overlap labels [4] and assumes future retain groups follow the calibration distribution; human adjudication improves test labels, but only 51 of 154 items had a full page available to at least one annotator, 103 decisions use snippets, and initial four-class agreement was modest before adjudication, so the human-label result audits the locked decisions rather than certifying human-clean population risk. External audits use lexical and constraint adapters on English and Chinese multiple-choice material; repeated-seed raw-search evaluation, stronger semantic baselines, code and task benchmarks, long documents, obfuscation, timestamped corpora, and additional languages remain untested.

VII. CONCLUSION

SafeDecontam converts retrieved benchmark-related evidence into source-level corpus-deletion decisions under a prespecified collateral-damage budget [8]. On the blinded adjudicated MMLU audit, the locked rule removes 67/82 contaminated and 0/71 clean source items, with a one-sided 95% exact clean-error upper bound of 4.13%; lower recall on external families and cross-lingual evidence shows that calibration cannot compensate for retrieval misses or evidence the adapter does not encode. The method is a calibrated deletion policy for retrieved candidates, not an end-to-end guarantee of

finding every contaminated record; broader use requires full-page annotation, difficult retain relations, and family-specific calibration data before deployment.

ACKNOWLEDGMENT

ChatGPT was used for language editing and LaTeX typesetting. The authors reviewed the manuscript, verified the reported results and citations, and take responsibility for the final text.

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